AGI ALPHA FLYWHEEL — When Capability Creates Opportunity

Evidence companion · Additive opportunity-discovery edition v4

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1. How to read the evidence

Observed means a source-processing, program, test, model-recipient or interface execution actually occurred in this workspace and has a receipt. It does not mean physical validation, customer acceptance or live identity authentication. **Sourced** means an attributable external statement or supplied artifact. **Assumed** means a researcher-selected premise. **Simulated** means a result of executing a model with those premises. No commercial quantity in this edition is calibrated to a customer cohort. Unknown cost or telemetry is explicitly null.

The headline result is intentionally split: an actual reusable instrument and fresh transfer exist; initial failures are retained; the operating-method trial shows no positive acceptance advantage; and modeled retention is rejected. There is no earned revenue, live part release or evidence of commercial compounding.

2. Custody and additive history

Item	Frozen source or path	Boundary
Research parent	`231834269599d85e49cfe1ca9917a1a874522d44`	Existing paper/correction history retained
Runtime source	`80b345828f34fd8a6e592977dd17ef37141305f4`	Existing bridge verifies its full vendor closure
Three foundational papers	`data/provenance/foundations.json`; `dossier/FOUNDATIONS_AUDIT.md`	Full public PDF/TeX reading; unseen attachments not presumed identical
Historical microgrid fixture	`../agi-alpha-economic-construction-v3/PRESERVATION.json`	Not this edition's flagship; failed/negative results remain
Opportunity budget and selection	`dossier/EXPLORATION_BUDGET.md`; `dossier/SELECTION.md`	Selection before new retained runtime construction
Current source coordination	GitHub issue #4850, comment 6447631102	Does not take over other research or release owners

The prior economic illustration retains USD300,000 successful contribution, USD120,000 expected contribution and USD900,000 prefunding under its supplied common-risk assumptions. The enterprise paper's 1,600-to-64 generic-case result is not a 96% total-work/customer-saving result; its supplied direct specialist remained faster in final timing. These corrections are not silently removed from the new narrative.

3. Opportunity-selection evidence

The dossier covers ten serious directions/challenges. Each records beneficiary, useful outcome, observed need versus conjecture, prior art, surviving gap, construction, decisive experiment, falsifier and commercial uncertainty. Ordinal scoring is researcher judgment, not statistical evidence. The actual research budget excludes paid API procurement, purchases, customer contact and hardware access; hosted-model cost and human effort are not observable.

Two small viability checks mattered. Thermal examples often became identifiable using established symmetry or construction information, so they did not support the proposed flagship. The runtime

pilot reproduced a reported older software failure but passed the same path on the authoritative current versions. Those negative findings were retained before the manufacturing direction was selected.

`dossier/runtime_pilot/` preserves the original bounded attempts, known-version controls and unresponsive initial attempts without inventing a causal explanation. No gain is credited to the team for an upstream software fix. Unused weather and reanalysis snapshots are marked unused; no scientific outcomes are fabricated.

The selected manufacturing gate required one complete source-to-geometry/result trace and a useful assertion beyond querying the supplied incoming graph. The Plate pilot met that construction gate. It did not establish demand, willingness to pay or superiority over existing commercial integrators.

4. Source data and rights

Asset	Bytes	SHA256 prefix	Status
Original NIST/MTC archive	248,727,407	`6bc4693f4b07`	Public historical experiment
NIST reference graph/model archive	10,680,120	`9ae303335274`	Supplied baseline graphs/models
Exact QIF 2.1 owner archive	See owner receipt	`d9e8b5f1333f`	Third-party schema/standard source
Exact Units.xsd	Bound in runtime manifest	`2d46993e4930`	Unmodified; DMSC notice accompanies distribution
Plate compact source pack	533,315	Bound in data receipt	Actually used development inputs
Original recipient public inputs	8,238,065	Per-file manifest	Separately provisioned, not transferred intelligence

The original archive's geometry is AP203; the reference archive's models are AP242. The Cover incoming original and reference copies differ in exact bytes. Source identity must bind the correct revision before graph qualification is reused. A byte mismatch alone does not prove a materially wrong interpretation.

NIST/MTC and reference notices accompany the data. DMSC's owner permission notice permits copying/use with the required copyright and permission notices; its schema is not relicensed as Apache or asserted to be public domain. The surrounding package includes `data/development/QIF_DMSC_NOTICE.md`, including for the immutable 0.1.0 bundle. Private customer material is absent. Public artifact access does not grant operational authority to manufacture or release a part.

5. Supplied ingredients versus constructed work

Supplied: the public manufacturing experiment, schemas, CAD engine, existing integrated graphs, known interval/permutation mathematics, ordinary tools, researcher-defined mission requests, evaluator criteria, model priors and finite commercial search menu.

Actually constructed: exact-source readers and reports; actual CAD candidates and traces; explicit unit/sign/identity findings; interval and assignment witnesses; a source-bound acquisition plan; a versioned operating method; an independent checker; an actual received-code invocation through existing interfaces; and complete hypothetical cash/accrual accounting.

Not established: autonomous buyer discovery, novel geometric algorithms, physical metrology, production admission, current ENS identity, learning to invent more efficiently, population reliability,

a new paying market or earned commercial Alpha. An inherited stronger artifact is not proof of an improved discovery policy.

6. Transformation witness and unit rule

The exact QIF 2.1 schema supplies the formal default: absent per-value and applicable primary linear units mean meters. Incoming files declare millimeters. The apparent plausibility of small dimensions cannot silently repair a source. A hypothetical millimeter interpretation is evaluated in a separate result.

For a common interval predicate, reordering the same selected records leaves its universal truth unchanged. The executable witness either checks every record or emits a concrete violating value. Interval inclusion separately checks whether the source requirement entails the destination requirement. Complete physical-family coverage, specimen relations, equivalent standards, measurement adequacy and current authority remain separate guards.

The five supported purposes are evidence audit, all selected recorded values, complete family, feature localization and physical release. Only the first two can sometimes be resolved without a physical bijection. A family certificate does not identify a hole; a local PASS does not release a part. The actual Plate trace contains a 3.017 mm diameter within its local interval and an overall FAIL in the parent result.

The record-only purpose was distinguished during development, before core freeze. The stronger already-curated requests retained their original purposes. No successful weaker statement was substituted for a failed stronger contract.

7. Frozen core and independent evaluation

Stage	Immutable identity / record	Result
Core 0.1.0 freeze	`instrument/freeze.json`; SHA `3526300d1499…`	Plate only, 11 focused tests
Generic runtime identity	`d25e9e0d1fea…`	Complete custom closure 116,202 bytes
Original evaluator freeze	`evaluation/EVALUATOR_FREEZE.json`; SHA `b5e165a9b447…`	19 adversarial Plate tests before fresh calls
Original eight-job confirmation	`evaluation/results/confirmation-v1/summary.json`	5/8 accepted scoped audits, no retries
Reserved recipient	`integration/results/confirmation-v1/summary.json`	1/1 accepted acquisition audit
Four prospective request freeze	`evaluation/prospective-v1/FREEZE.json`; SHA `9bb07729b6c2…`	Before core repair
Separate core 0.1.1	`instrument_v0_1_1/freeze.json`; SHA `107338d3358d…`	13 Plate-only tests
Separate prospective evaluator	`evaluation/prospective-v1/PROSPECTIVE_EVALUATOR_FREEZE.json`	Six targeted adapter checks before calls
Four prospective results	`evaluation/prospective-v1/results/confirmation-v1/summary.json`	4/4 new requests, no retries

The original three failures are general Box, general Cover and Cover without CAD. Each omitted affected-object bindings in two repair requests. Their mathematical/geometry claims passing did not make the whole requested report acceptable. Original compressed reports preserve exact bytes. The successful repair is not relabeled as original performance or independent physical data.

The checker independently parses XML and Decimal quantities and directly reimports CAD through

OCP, without importing candidate proof functions. Candidate and checker still share the underlying geometry kernel. Protection was explicit agent-context/file-access discipline, not a hostile-host sandbox. The data are public and from one assembly. Neither thousands of reference checks nor multiple replays are independent examples.

Fresh source conclusions

Box selected source characteristics 14/20/26/32 provide 76 values. Under the explicit millimeter hypothesis the destination interval is [3.1, 3.5] mm and the observed range is [3.246, 3.536] mm. Actual 988 in result 984 records 3.506 mm, a concrete counterexample. Source PASS under a wider interval does not suffice.

Cover's twenty nominal-4.5 source characteristics provide 400 values. Under the hypothesis they range from 4.523 to 4.537 mm inside [4.2, 4.8] mm. The numeric record statement is supported. The receiving family represents four feature identifiers; the source collection does not establish the required physical coverage. Localization and release remain unsupported.

8. Fresh transfer and authority

The actual transport ZIP is 43,242 bytes, SHA256 `db571361b945bffc46c602713c4fa09d6e667ee9a8dc2d54c46dfa42f1784bd0`. It passes through the unchanged source-verified bridge. The installed dependency closure is separately provisioned and rehashed; it is not squeezed into the small artifact or called transferred intelligence.

Thirty-five study controls and twenty-nine inherited lifecycle controls pass. They include altered bytes, current-source changes, dependency changes, expired or revoked signatures, a changed mission and consumed-grant replay. Six generic integrity cases can be cached under exact identity, while current source, semantic and authority work remains. This is not a 100% reduction in total work.

The local research grant uses native Ed25519 roles and Journal nonce recording. Modeled agent names are not live ENS identities. The application reads public sources with before/after hash checks; this is not an OS-enforced read-only mount. The native capsule's refusal of unsupported manufacturing scope is preserved. Jobs validators, governance, settlement and revocation rules are unchanged.

The ordinary library has identical content and permissions, and its report has semantic parity. No graph-specific improvement is established. Service access, executable workflow transfer and independent ownership of specialist weights are not conflated.

9. Cheap Operational Alpha experiment

`expertise/freeze.json` binds the four-job packet, rubric, protocol and method before recipient dispatch. H remains estimated. U is competent ordinary use, E adds an established expert procedure, and E+ adds the retained method only; the larger instrument is not supplied as an extra shortcut. All arms have the same exact sources, useful selectors, ordinary tools and exact unit schema.

Arm	Strict jobs	Criteria	Calls, self-reported	Words, counted
U	4/4	24/24	7	1,239
E	4/4	24/24	6	1,185
E+	3/4	23/24	8	1,207

The strict E+ failure is an inconsistent component/serial instruction in H4. Its correct earlier trace is retained, and a lenient 4/4 sensitivity is disclosed. Neither reading demonstrates additional accepted-quality Alpha. One execution error was corrected within E+'s common budget. This is not evidence of general method-caused harm.

The eight-call and 1,600-word ceilings match observable allowances, not exact inference expen-

diture. Token/billing telemetry, active latency and real customer effort are unavailable. Cost at matched accepted quality is therefore unknown. The grading is role-separated AI-assisted judgment under a frozen rubric, not blinded human metrology. No optional Gym or weight-training campaign was run.

10. Economic assumptions, freeze and complete ledger

`economics/PROTOCOL.md`, `model.py`, `trajectory.py` and `admission.json` are bound by `economics/freeze.json`. Development seeds are 110–121; primary confirmation seeds are 21000–21079; forced lineage diagnostics use 22000–22119. Future-event uniforms are generated after package selection, outside planner inputs.

Every price, demand, value, reliability, cash labor rate, support cost and risk parameter is assumed. None is calibrated from the small source trial. Human service is an estimated team-hour/rate reference, with an efficient configured recipient control. Prices, buyer subsets and commission terms are searched; no fixed discount or coalition is mandatory. The model can stop.

A is complete paid-job work, including required common construction and failure costs. B is additional retention/upkeep. C is additional research/assessment, including rejected assessments. In the separate trajectory, initial optional construction is C once and transfer is B. Actual unpaid session research remains C-like research with unknown full cost; its expense is not silently assigned the model's USD100 prospecting assumption.

Collections, unearned advances, refunds, reserves, invoices, bad debt, expenses, profit and seed are separate. Native CostLedger receives a settled projection, while the full dated research ledger preserves obligations and timing. The selected native replay passes twelve conformance checks. Its first floating- point signing refusal and decimal-string correction remain in history.

The frozen model conditions financing on acceptance of the development record. It does not simulate rejection of that milestone, a legal dispute, dishonest execution or every insolvency process. A separate later audit freezes and tests full rejection/refund exposure without changing primary outcomes. It finds 114 of 1,889 selected commitment plans exposed; eleven of those were subsequently cancelled in the original withdrawal paths. These are contingent exposures, not realized defaults or customers. Original full-refund-term selections add no exposure under that stress.

11. Results that do not cross the frontier

The primary simulation comprises 3,520 arm-horizons. Central financeable counts are 74/80 expert standard, 27/80 retained standard, 75/80 expert arranged and 38/80 retained arranged. All eleven stresses show zero combined-versus-expert- standard feasibility crossings. Both qualifying is not a crossing.

The six-horizon admission calculation rejects USD2,950 additional investment against USD2,220.94 modeled benefit. Forced construction yields mean final incremental value −USD1,049.79, with 2/20 nonnegative paths. Its rejection is not retrospectively reversed. Identical-content library economics are identical.

No physical-release frontier, independently measured human-service saving, positive learned-method advantage, paid customer base, token valuation or commercial compounding has been established. Profitable expert delivery remains a conditional economic result, independent of rejected optional learning.

12. Reproducibility and residual limits

The study README gives the commands, exact source paths, dependency versions and hash verification. Compact input packs and restoration scripts preserve original bytes. Replaying the released fixture is reproduction, not another fresh confirmation. New claims require new protected needs and acceptance criteria.

The full repository software/release suite was not executed. Targeted runtime, evaluator, integration and economic checks are recorded individually. Ruff format/check was used; installed Black had a `pathspec` import problem. No production release or canonical Pages qualification is inferred from these checks.

The browser demo replays frozen evidence and separately labeled financial assumptions; it does not call a frontier model or authenticate a live network. The original vision motivates the research, but its market/IP superlatives are not accepted findings or forecast inputs. The exact title and author attribution are supplied by the commissioning user; AI assistance and source provenance are disclosed. Current public draft links and publication status are recorded in the study README and publication receipt.